

video content or color grading HDR). When checking in store, note that some glossy screens (especially OLED) are harder to see in bright light despite high contrast, so raw brightness matters if you plan to use the laptop outdoors. Mini-LED and some IPS can sustain higher full-screen brightness than OLED – for instance, the MacBook Pro’s mini-LED hits ~500 nits sustained full-screen. If outdoor usability is key, consider an anti-reflective coating or matte screen option if available, and at least ~400 nits. Many business laptops (ThinkPads, Latitudes) offer optional high-brightness panels (500 nits, sometimes with anti-glare even on touch versions).

- **Colour Gamut & Accuracy:** This matters if you do any color-sensitive work (photo editing, video grading, design) or if you just care about vibrant visuals. Standard gamut is sRGB (~100%) which most panels can cover. Wider gamuts include Adobe RGB and DCI-P3. Many premium screens target 100% DCI-P3 (common for OLED and some high-end IPS). A few specialized models cover near 100% Adobe RGB (useful for print photography folks) – often these are labeled as “DreamColor”, “TruLife”, or come factory-calibrated. If this is important, check reviews for gamut coverage. For average users: as long as it’s close to 100% sRGB, you’ll get a good colour. Sidenote: colour accuracy (Delta-E) is rarely listed in specs, but reviews might mention it. If you see a laptop marketed as calibrated (like XPS 15, certain ASUS ProArt or Razer Studio editions), it means out-of-the-box colors are tuned well. For designers, that’s a plus. Otherwise, you can always calibrate later with a colorimeter if needed.
- **Avoiding eye strain – PWM and blue light:** As mentioned, OLEDs often use PWM (pulse-width modulation) to adjust brightness. Essentially the display flickers faster than your eye can see – but your brain might still detect it, causing fatigue. Some IPS panels use PWM too, but many do not or only at very low brightness. If you are sensitive, try to find flicker-free displays (some companies advertise “DC dimming” or certify as Eyesafe). You can test for PWM by pointing your phone’s camera at the screen at low brightness – if you see lines or a strobe effect through your camera, there’s PWM. High-frequency PWM (>20 KHz) is less problematic. Additionally, many laptops now have blue light reduction modes or claim “low blue light” certified panels – helpful for late-night use to reduce eye strain (without severely skewing colors). It’s a nice feature but not a make-or-break; software night modes can do similar.

5. Ports & Expansion: USB4, Thunderbolt, HDMI

Modern laptops can come with an array of ports, or practically none at all (looking at you, super-slim ultrabooks). It’s crucial to ensure the laptop has the connectivity you require for peripherals, displays, and future expansion. Here are the key ports and what they mean, and guidance on who needs what:



- **USB Type-C / USB4 / Thunderbolt:** Today, USB-C is the universal port for charging and connectivity. However, not all USB-C ports are equal. Some support Thunderbolt 4 or USB4, which means up to 40 Gbps bandwidth, the ability to drive multiple 4K monitors, and support for eGPUs and high-speed docks. Others might be USB 3.2 Gen2 (10 Gbps) or even just Gen1 (5 Gbps) with only data and maybe DisplayPort out. Thunderbolt 4 (found on many Intel-based and some AMD laptops via USB4) guarantees certain capabilities: 40 Gbps, at least two 4K displays output, and PCIe support. Thunderbolt 5 (the next-gen, we’ll cover soon) doubles this. If you are a power user who connects to multi-monitor setups or uses external drives/docks, make sure at least one of the USB-C ports is Thunderbolt/USB4. For instance, a Dell XPS 15 might come with 2x Thunderbolt 4 USB-C ports – allowing you to plug in one cable to a dock that handles charging, dual monitors, and peripherals. On the other hand, a budget thin-and-light might have a physically identical USB-C port that only does data + charging, but no video output – that could be a problem if you plan to connect an external monitor via USB-C (you’d need a separate HDMI in that case). Always check the specs or markings: Thunderbolt ports often have a little lightning bolt symbol. USB-C with DisplayPort Alternate Mode (for video out) might have a DisplayPort “DP” logo. If unsure, read the laptop’s spec sheet to see if it mentions external display support via USB-C.
- **USB-A (the classic rectangular port):** Many laptops still include at least one USB-A for connecting older devices (flash drives, mice, keyboards, etc.). Some super-thin ones have dropped USB-A entirely – be prepared to live the dongle life if so. If you have a lot of legacy peripherals, a laptop with 2x USB-A is convenient (e.g. many Lenovo ThinkPads and HP Pavillions include two). For others, one is enough (you can always use a hub if needed). USB-A ports also have versions (3.2 Gen1 = 5 Gbps, Gen2 = 10 Gbps, etc.), but for typical use (mice, printers, thumb drives) the speed isn’t critical. If you frequently transfer files on external SSDs, you’ll be using either a USB-C or a fast USB-A (marked in blue or red usually).